WORKING TITLE

by

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Abstract

My thesis is about bla bla

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Thanks!

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Chapter 2

Background

2.1 Multicategories

EXPLAIN THE SOURCE OF THIS MATERIAL. "Multicategories were first defined in ..." "Our exposition follows ..." yes Multicategories are a generalization of categories. In categories, arrows have a single source/domain and a single target/codomain. In multicategories, the arrows have a finite sequence of sources/inputs and a single target/output.

Definition 2.1.1. A *multicategory* $\mathcal{M}$ consist of a set S of objects and for each $n \geq 0$ and each sequence $s_1, \ldots, s_n, t$ of objects a set

$$\mathcal{M}(s_1, \ldots, s_n; t)$$

of multiarrows/operations, thought of as taking n inputs objects $s_1, \ldots, s_n$ respectively to the output object t. Moreover, there are three kinds of structure maps:

- For each object s, an identity/unit $1_s \in \mathcal{M}(s; s)$

- For any sequence of objects $s_1, \ldots, s_n, t$ and any n-tuple of sequences $d_1^i, \ldots, d_{k_i}^i$ of objects for $i = 1, \ldots, n$, a composition function of multiarrows/operations

$$\gamma : \mathcal{M}(s_1, \ldots, s_n; t) \times \prod_{i=1}^{n} \mathcal{M}(d_1^i, \ldots, d_{k_i}^i; s_i) \to \mathcal{M}(d_1^1, \ldots, d_{k_1}^1, d_1^2, \ldots, d_{k_n}^n; t),$$

usually written $p \circ (q_1, \ldots, q_n) := \gamma(p, q_1, \ldots, q_n)$ or even just $p(q_1, \ldots, q_n)$

Furthermore these structure maps have to satisfy a number of axioms, which express that composition is unital and associative (Heuts and Moerdijk, 2022)

Definition 2.1.2. A (colored) operad $\boldsymbol{P}$ is a multicategory with a symmetric action on the set of multiarrows. That is:

- For each permutation $\sigma \in \Sigma_n$, a map

$$\sigma^* : \boldsymbol{P}(c_1, \ldots, c_n; c) \to \boldsymbol{P}(c_{\sigma(1)}, \ldots, c_{\sigma(n)}; c)$$

usually written $\sigma^* p = p \circ \sigma$

such that these maps are compatible with the composition and the units giving a right action of the symmetric groups on the sets of multiarrows.

We will be calling multicategories when we are not interested in the symmetric action, and operads when we are. In this thesis we will be mainly interested in multicategories, but we will also talk about operads at some points.

Remark 2.1.3. Let $\mathcal{M}$ be a multicategory. Given multiarrows $f \in \mathcal{M}(s_1, \ldots, s_n; t)$ and $g \in \mathcal{M}(d_1, \ldots, d_l; s_i)$ for some $1 \leq i \leq n$. It will be convenient to have the following notation:

$$f \circ_i g := f(1_{s_1}, \ldots, g, \ldots, 1_{s_n})$$

Thus, $f \circ_i g$ is the composition of f with g in the ith variable.

Example 2.1.4. You can think of n-ary multiarrows in $\mathcal{M}(s_1, \ldots, s_n; t)$ as n-ary operations. Let M be the multicategory with a single object X. Then the n-ary multiarrows in $M(X, \ldots, X; X)$define to be the set of functions

$$f : X \times \cdots \times X \to X$$

where the domain is the n-fold cartesian product of X with itself.

Example 2.1.5. We can up the level from the previous example, by fixing a family of sets $\{X_c\}_{c \in C}$ indexed by a set of objects C. Then you can think of the multiarrows in $\mathcal{M}(c_1, \ldots, c_n; c)$ as n-ary operations, taking inputs of types $c_1, \ldots, c_n$ respectively to an output of type c.

For example, for a fixed family of sets $\{X_c\}_{c \in C}$, define a multicategory $\mathcal{M}$ whose objects is the set of colors C, and whose multiarrows are defined by declaring $\mathcal{M}(c_1, \ldots, c_n; c)$ to be the set of functions

$$p : X_{c_1} \times \cdots \times X_{c_n} \to X_c$$

Composition in $\mathcal{M}$ is defined as composition of functions.

Example 2.1.6. The previous two examples are specific instances of a more general family of examples example. If $\mathcal{C}$ is a symmetric monoidal category and C is a set of objects of $\mathcal{C}$, then one obtains a multicategory with this set of objects by defining $\mathcal{M}(c_1, \ldots, c_m; t)$ to be the set of morphisms $c_1 \otimes \cdots \otimes c_n \to$

t in $\mathcal{C}$. We will sometimes write $\mathcal{C}^{\otimes}$ for the multicategory obtained in this way, when C is the set of all objects of $\mathcal{C}$ (granted it is small).

Example 2.1.7. A (small) category $\mathcal{C}$ can be considered as a multicategory $\mathcal{C}_m$ which only have unary arrows. I.e. the set $\mathcal{C}_m(c_1, \ldots, c_n; c)$ is always empty unless $n = 1$.

We can observe that multicategories form a category $\mathcal{MC}\mathsf{at}$.

Definition 2.1.8. For two multicategories $\mathcal{M}$ and $\mathcal{N}$, a *multifunctor* $\phi : \mathcal{M} \to \mathcal{N}$ consist of:

- A function $f : O_{\mathcal{M}} \to O_{\mathcal{N}}$ between the corresponding sets of objects
- For each sequence $s_1, \ldots, s_n, t$ of objects of $\mathcal{M}$, a map

$$\phi_{s_1,\ldots,s_n,t} : \mathcal{M}(s_1, \ldots, s_n; t) \to \mathcal{N}(f(s_1), \ldots, f(s_n); f(t)).$$

So that these maps are compatible with compositions and units in the obvious way.

2.1.1 The Tensor Product of Operads

Let M and N be operads. Their Boardman-Vogt tensor product $M \otimes_{BV} N$ is a operad defined to make sense of what a M-algebra in the category of N-algebras is and vice versa. We will not talk about the algebras of a multicategories here, but we refer the reader to [look for reference]

Definition 2.1.9. We define the *Boardman-Vogt tensor* $M \otimes_{BV} N$ in terms of generators and relations. The set of objects of this multicategory is the product of the sets of objects of M and N; if x and y are objects of M and N respectively, we write $x \otimes y$ for the corresponding object of $M \otimes_{BV} N$. The generating multiarrows of $M \otimes_{BV} N$ are of two types:

- For each multiarrow $f \in M(x_1, \ldots, x_n, x)$ and object y of N, a multiarrow

$$f \otimes y \in (M \otimes_{BV} N)(x_1 \otimes y, \ldots, x_n \otimes y; x \otimes y)$$

- For each object x of M and multiarrow $g \in N(y_1, \ldots, y_m; y)$, a multiarrow

$$x \otimes g \in (M \otimes_{BV} N)(x \otimes y_1, \ldots, x \otimes y_m; x \otimes y)$$

The relations to be satisfied by these generators are the following:

(1) $(f \otimes y)(f_1 \otimes y, \ldots, f_n \otimes y) = f(f_1, \ldots, f_n) \otimes y$

(2) $(x \otimes g)(x \otimes g_1, \ldots, x \otimes g_m) = x \otimes g(g_1, \ldots, g_m)$

(3) $\sigma^*(f \otimes y) = (\sigma^* f) \otimes y$ for $\sigma \in \Sigma_n$

(4) $\sigma^*(x \otimes g) = x \otimes (\sigma^* g)$ for $\sigma \in \Sigma_m$

(5) $(f \otimes y)(x_1 \otimes g, \ldots, x_n \otimes g) = \sigma^*_{n,m}((x \otimes g)(f \otimes y_1, \ldots, f \otimes y_m))$

Note that relations (1) and (2) say precisely the condition that for any object y of N there is a multifunction $- \otimes y : M \to M \otimes_{BV} N$. Similarly, for any object x of M there is a multifunction $x \otimes - : N \to M \otimes_{BV} N$, this time given by relations (2) and (4). Relation (5) is the most interesting; most often referred to it as the *Boardman-Vogt interchange relation*. The permutation $\sigma_{n,m} \in \Sigma_{nm}$ is the one that makes sense of formula (5), i.e. the element relating the sequences $(x_1 \otimes y_1, \ldots, x_1 \otimes y_m, \ldots, x_n \otimes y_1, \ldots, x_n \otimes y_m)$ and $(x_1 \otimes d_1, \ldots, x_n \otimes d_1, \ldots, x_1 \otimes d_m, \ldots, x_n \otimes d_m)$.

2.2 Trees

We will do a quick detour to talk about our main combinatorial object that will help us with studying multicategories: trees. We are mainly interested in finite rooted trees. So we don't get confused with the more general notion of tree in graph theory, we will give a precise definition of what we mean by a tree in this thesis. We will see that these trees give rise to free multicategories whose objects are the edges of the tree and morphisms are generated by the vertices of the tree.

Definition 2.2.1. A *tree* T consist of a finite set V of *vertices*, a nonempty finite set E of *edges*, a distinguished element $r \in E$ called the *root*, together with the following data:

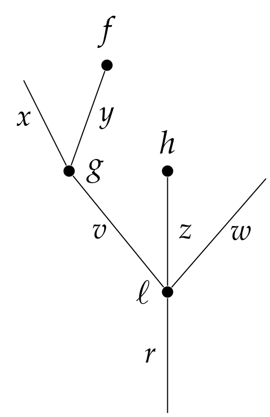
(1) A function $I : E - \{r\} \to V$, which we think of as assigning to an edge e the vertex $I(e)$ of which is an input. The way to think about this function is that "$I(e) = f$ if and only if e is *an* input edge of f"

(2) An injective function $O : V \to E$, assigning to each vertex g its output edge $O(g)$. The way to think about this function is that "$O(f) = x$ if and only if x is *the* output edge of f"

For each edge e other than the root r, we obtain a sequence of edges starting at e by repeatedly applying $O \circ I$. We demand that this sequence ends in the root r after finitely many steps, for an arbitrary starting edge.

As we can see the main difference between our definition of a tree and the usual definition of a tree in graph theory is that we allow for "orphan"/"leaf" edges that are not attached to any vertex as an output edge, and a unique designed edge that will not be attached to any vertex as an input edge (the root). This special edge also gives us a natural orientation of the tree, where we think of the root as the bottom and the leaves as the top, and thus a partial ordering on both the set of edges and the set of vertices.

Example 2.2.2. A typical tree T with $E(T) = \{x, y, z, w, v, r\}$ and $V(T) = \{f, g, h, l\}$ could look like:



We can fully define both functions for this small example as:

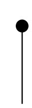
$$I(x) = I(y) = g, \;\; I(v) = I(z) = I(w) = \ell$$

$$O(f) = y, \;\; O(g) = v, \;\; O(h) = z, \;\; O(\ell) = r$$

Now I will introduce some terminology to talk about the different parts of a tree. The edges in the complement of the image of O are called *leaves* of the tree T. The vertices in the complement of the image of I are called *stumps*, or *nullary vertices.* An *outer edge* is an edge that is either the root or a leaf. An *inner edge* is any other edge of T. Such an edge is both an output edge and an input edge for some other vertex. The vertices come in two kinds, *external vertices* which only have one inner edge attached to them, and *internal vertices* When writing T for a tree, we will usually write $E(T)$ and $V(T)$ for its sets of edges and vertices, respectively.

Example 2.2.3. The smallest possible tree consists of a single edge and no vertices; this tree will be denoted by μ

Example 2.2.4. The next smallest tree consist of a single edge and single vertex, pictured as:



The valence of a vertex v is the number of input edges it has, i.e. the cardinality of the set $I^{-1}(v)$. A vertex of valence 0 is what we previously called a stump. We previously mentioned that the sets $E(T)$ and $V(T)$ both have a natural partial ordering. Expounding it, it means that $x < y$ for edges x and y (or $f < g$ for vertices f and g) if there is a sequence of edges starting at x (resp. f) and ending at the root of T, r, and passes through y (resp. g). We will call two edges x and y *incomparable* if they are not related in this partial order.

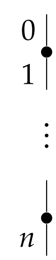
We note that removing an external vertex and all the outer edges attached to it yields a new tree. We will refer to this procedura as *pruning*.

Definition 2.2.5. A tree is called *open* if it has no nullary vertices.

Definition 2.2.6. A tree is called *closed* if it has no leaves.

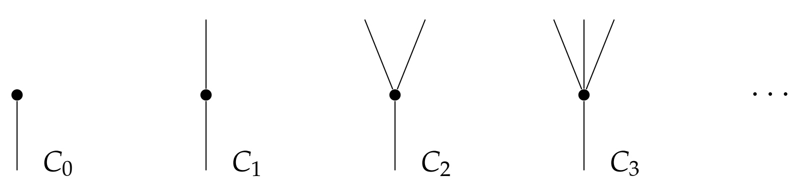
Definition 2.2.7. A tree is called *linear* if all its vertices have valence one.

As an example, the tree μ is open but not closed, and generally a tree cannot be both open and closed. A linear tree with $n+1$ edges running from the leaf 0 to the root n will be denoted by $[n]$:



The reason for this notation is that the linear trees will become clear in the following section. Note that by our definition the tree obtained from $[n]$ by putting a stump on top is not a linear tree since the stump has valence zero..

Another family of trees that we will become familiar with are the *corollas*, which are trees with a single vertex. The corolla with n leaves labeled $1, \dots n$ and root 0 will be denoted by C_n:



Definition 2.2.8. An *embedding* $\psi : S \to T$ of trees consists of injective maps $\psi_V : V(S) \to V(T)$ and $\psi_E : E(S) \to E(T)$ such that for any vertex f of S, the map ψ_E gives a bijection between $O^{-1}(f)$ and $O_{-1}(\psi_V(f))$, and $\psi_E(O(f)) = O(\psi_V(f))$. If an embedding is bijective on both edges and vertices, then it is an isomorphism of trees.

If an embedding of trees $\iota : S \to T$ is given by inclusion maps of $E(S) \subseteq E(T)$ and $V(S) \subseteq V(T)$, we will say that S is a *subtree* of T. Then we can see that any embedding $\psi : S \to T$ factors as an isomorphism followed by the inclusion of a subtree.

Definition 2.2.9. A tree T defines a multicategory $\mathbf{\Omega}(T)$ as follows:

- The objects of $\mathbf{\Omega}(T)$ are the edges of T.
- $\mathbf{\Omega}(T)(e_1, \ldots, e_n; e)$ is either empty or a singleton if there exist a subtree C_n with leaves $e_1, \ldots, e_n$ and root e.
- Composition is given by the tree structure in the obvious way. Expanding the details of it, we get the composition operation/multiarrow by identifying the root of the precomposing operation with a leave of the postcomposing operation and squashing this new inner edge and identifying the two vertices to the composition of the two corresponding operations/multiarrows.

2.3 Category of trees $\mathbf{\Omega}$

Dendroidal sets are a generalization of simplicial sets that are particularly well suited to study operads/multicategories and multicategories. Just as simplicial sets are presheaves on the simplex category $\mathbf{\Delta}$, dendroidal sets are presheaves on the dendroidal category $\mathbf{\Omega}$.

Definition 2.3.1. The category $\mathbf{\Omega}$ consist of:

1. Objects: finite rooted trees

2. Morphisms: for two trees S and T, a morphism from S to T is a multi-functor $\mathbf{\Omega}(S) \to \mathbf{\Omega}(T)$ between the corresponding multicategories.

Another way to say this is that $\mathbf{\Omega}$ is the full subcategory of $\mathcal{MC}\mathsf{at}$ spanned by the multicategories of the form $\mathbf{\Omega}(T)$ for some tree T.

The linear trees $[n]$ define a full subcategory of $\mathbf{\Omega}$ isomorphic to the simplex category $\mathbf{\Delta}$. Thus, there is an inclusion functor $i : \mathbf{\Delta} \to \mathbf{\Omega}$. This inclusion is compatible with the inclusion of Cat into $\mathcal{MC}\mathsf{at}$ in the sense that the following diagram commutes:

$$\begin{tikzcd} \mathbf{\Delta} \arrow[r] \arrow[d, "i"'] & \mathsf{Cat} \arrow[d, "j"] \\ \mathbf{\Omega} \arrow[r] & \mathcal{MC}\mathsf{at} \end{tikzcd}$$

2.4 Faces and Degeneracies in $\mathbf{\Omega}$

As we know, the simplex category $\mathbf{\Delta}$ has two important families of morphisms, the face maps and the degeneracy maps. These maps generate all other morphisms in $\mathbf{\Delta}$. The good news for $\mathbf{\Omega}$ being some multicategorical extension of $\mathbf{\Delta}$ is that it also has two families of morphisms that generate all other morphisms in $\mathbf{\Omega}$.

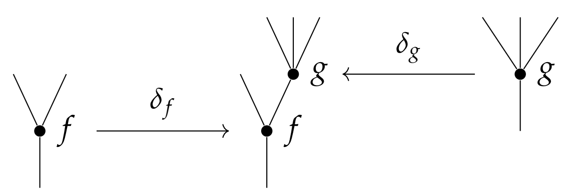
2.4.1 Outer Faces

For a tree T and an external vertex f of T, write $\partial_f T$ for the tree obtained by pruning away f and all the outer edges attached to it. This vertex could be any leaf vertex or it can be the vertex attached to the root of T, provided it is only attached to one inner vertex of T. We write

$$\partial_f T \xrightarrow{\delta_f} T$$

for the inclusion of this tree and refer to it as an *outer face*. If f is a leaf vertex we will sometimes specifically speak of a *leaf face*, and similarly for a *root* face.

Example 2.4.1. A leaf face $\partial_g T$ given by the vertex g and a root face $\partial_f T$ given by the vertex f of the following tree T:



Remark 2.4.2. Corollas are a special case for the outer face maps. Corollas, by our definition, are trees with only one internal vertex and n leaves. However, for corollas we have $n+1$ different maps $\mu \to C_n$ corresponding to the inclusion of each of the n leaves and the root. We define all of these to be outer faces as well.

2.4.2 Inner Faces

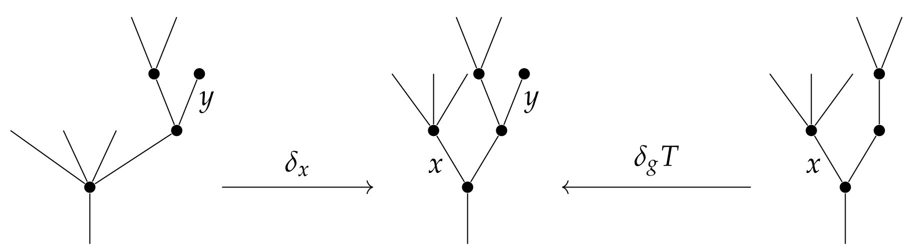
For a tree T and an inner edge x, let $\partial_x T$ denote the face of T obtained by contracting the edge x, and identifying the vertices f and g at either end of x.

This is an *inner face* of T. This gives an inclusion of trees denoted

$$\partial_x T \xrightarrow{\delta_x} T$$

which sends the new vertex obtained by contraction to the subtree of T obtained by grafting the two corollas with vertices f and g along the edge x, i.e. the smallest subtree of T containing these vertices. We can see these in terms of their associated operads. In this framework, we see that we send the new vertex in $\partial_e T$ to the composition in $\mathbf{\Omega}(T)$ of the operations correspondingly given by f and g along x. A morphism of these kind will be referred to as an *inner face*

Example 2.4.3. The inner face $\partial_x T$ and $\partial_y T$ given by the edges x and y of the following tree T:



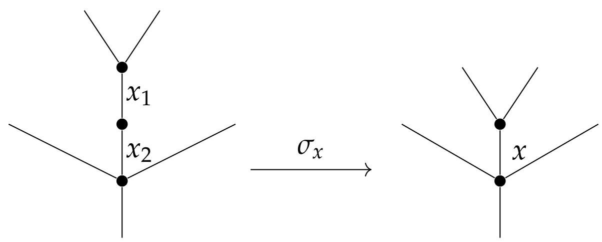
2.4.3 Degeneracies

Let x be an edge in a tree T. Consider the tree $\sigma_e T$ obtained from T by splitting this edge into two edges x_1 and x_2 by placing an unary vertex id_x in the middle of x. The there is a morphism

$$\sigma_x T \xrightarrow{\sigma_e} T$$

in $\mathbf{\Omega}$ which sends both edges x_1 and x_2 to x and the new unary vertex id_x to the subtree $\mu \xrightarrow{x} T$ of T. In the operadic framework, we see that it sends the new vertex id_x to the identity operation of the color x in $\mathbf{\Omega}(T)$. A morphism of this kind is called a *degeneracy*.

Example 2.4.4. The degeneracy $\sigma_x T$ given by the edge x of the following tree T:



As with simplices, we have codendroidal identities. We won't go over them since we won't be using them in any kind of way. To see a full list and more on them please refer to Heuts and Moerdijk, 2022, §3.3.4. Following suit from simplices, we can also prove that any morphism of trees can be factorized into a composition of a degeneracy, followed by an isomorphism and ending with a face map.

2.5 Dendroidal Sets

As no surprise to the reader, the definition of a *dendroidal set* is analogous to the definition of a simplicial set.

Definition 2.5.1. A dendroidal set is a functor

$$X : \mathbf{\Omega}^{\mathrm{op}} \to \mathcal{S}\mathsf{et}$$

Natural transformations as morphism between them, makes them form the category of dendroidal set, d$\mathcal{S}$et

Expounding out the definition of a dendroidal sets, we can see a dendroidal set is given by a family of sets X_T, one for each $T \in \mathbf{\Omega}$, together with a map

$$\alpha^* \; : \; X_T \to X_S$$

for every morphism $\alpha : S \to t$ in $\mathbf{\Omega}$. These maps should be functorial. The elements of X_T are called *dendrices* of X of shape T, or simply T-dendrices of X.

In particular, each set X_T carries a n action of the group $Aut(T)$ and the dendroidal set X is completely determined by the collection of all these $Aut(T)$-sets together with the elementary face and degeneracy operators of the three kinds we saw before, satisfying the *dendroidal identities*. dual to the identities previously mentioned.

Example 2.5.2. Any tree T of $\mathbf{\Omega}$ gives rise to a representable presheaf which we denote by $\mathbf{\Omega}[T]$. As in any presheaf category this are defined by

$$\mathbf{\Omega}[T]_S = \mathrm{Hom}_{\mathbf{\Omega}}(S, T)$$

Example 2.5.3. One can from limits and colimits of diagrams of dendroidal sets by calculating them 'pointwise', as in any presheaf category. For instance, we can from the product $X \times Y$ of dendroidal sets X and Y as

$$(X \times Y)_T = X_T \times Y_T$$

Example 2.5.4. In particular, each face $\delta_x : \partial_x T \to T$, where x is an inner edge or outer vertex, defines a monomorphism of dendroidal sets

$$\mathbf{\Omega}[\partial_x T] \to \mathbf{\Omega}[T]$$

. We will write $\partial_c \mathbf{\Omega}[T] \subseteq \mathbf{\Omega}[T]$ for the image of this morphism and call it the *face of T opposite c*. The union of all these (inner and outer faces) defines the *boundary* $\partial\mathbf{\Omega}[T]$ of $\mathbf{\Omega}[T]$. Note that these faces and the boundary can be explicitly described as follows:

$$(\partial_c \mathbf{\Omega}[T])_S = \{\alpha : S \to T \mid \text{the edge } c \text{ is not contained in the image of } \alpha\},$$

$$(\partial_f \mathbf{\Omega}[T])_S = \{\alpha : S \to T \mid \text{no subtree } \alpha(g) \text{ contains the vertex } f\},$$

$$(\partial \mathbf{\Omega}[T])_S = \{\alpha : S \to T \mid \alpha \text{ factors through a proper inclusion } R \to T\}.$$

Example 2.5.5. The category $\mathbf{\Omega}$ has been defined as a full subcategory of the category in $\mathcal{S}$et. Thus any such operad $\boldsymbol{M}$ defines a dendroidal set N$\boldsymbol{M}$, its *dendroidal nerve*, by

$$\mathrm{N}\boldsymbol{M}_T = \boldsymbol{Op}(\mathbf{\Omega}(T), \boldsymbol{M}).$$

Even though this is an standard construction in homotopical category theory, let us take a closer look at the dendroidal set given by the operad. For the tree μ, the set N$\boldsymbol{M}_\mu$ is the set of colors of $\boldsymbol{M}$.For the n-corolla C_n and a sequence of colors $(c_1, \ldots, c_n; c)$ a sequence of colors of $\boldsymbol{M}$, there is a pullback square

$$\begin{tikzcd}
\boldsymbol{M}(c_1, \ldots, c_n; c) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{N}\boldsymbol{M}_{C_n} \arrow[d] \\
* \arrow[r] & \prod_{i=0}^{n} \mathrm{N}\boldsymbol{M}_\mu
\end{tikzcd}$$

Here the right vertical map has as its i^{th} component the map $i^* : \mathrm{N}\boldsymbol{M}_{C_n} \to \mathrm{N}\boldsymbol{M}_\mu$, induced by the edge inclusion $i : \mu \to C_n$, while $\boldsymbol{c}$ denotes the sequence $(c_1, \ldots, c_n; c)$. More generally, for a tree T a dendrex in $\mathrm{N}\boldsymbol{M}_T$ can be pictured as follows. It consist of an assignments x of colors of $\boldsymbol{M}$ to the edges of T and a equivalence class of pairs (p, f), where p is a planar structure on T and f assigns an operation $f(v) \in \boldsymbol{M}(c(e_1), \ldots, c(e_n); c(e))$ to each vertex $v \in T$, where e denotes the output edge of v and $e_1, \ldots, e_n$ enumerates the input edges of v in the order provided by the planar structure p. Two such pairs (p, f) and (q, g) are equivalent if for each vertex v, the permutation σ relating the planar structure p and q satisfies $f(v) = \sigma^* g(v)$.

A little more informally, a dendrex in $\mathrm{N}\boldsymbol{M}_T$ is an assignment of colors of $\boldsymbol{M}$ to the edges of T and a compatible assignment of an operation in $\boldsymbol{M}$ to each vertex of T. As explained above, this is only a precise statement if one has a planar structure on T in minds.

Example 2.5.6.

Bibliography

Heuts, Gijs and Ieke Moerdijk (2022). *Simplicial and Dendroidal Homotopy Theory*. Springer Cham.